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RESEARCH PAPER

Time to break up? The case for tailor-made digital platform regulation based on platform-governance standard types

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Abstract
The rise of hybrid-conglomerate platforms like Google, Apple, Microsoft, Amazon, and Meta (GAMAM) has led to significant market power concentration and negative competition implications, right down to calls for their potential breakups. To expand dominance, these platforms leverage extensive ecosystems with strategic control over resources and customer relationships, often at the expense of others. We conduct a qualitative meta-analysis of 87 empirical platform-governance problem cases from a regulatory perspective. Using a decomposition approach, we develop a taxonomy of competition problems induced by platform governance and identify four governance standard types (i.e., growth, consolidation, extension, and protect and capture), representing platform lifecycle phases from a regulatory perspective. Each standard type describes the platform owner's governance strategy, consequent competition problems, and potential remedies. Our findings show how lack of regulation of digital platforms' governance design has contributed to their growing market power and that effective regulatory intervention requires making platform governance more open and neutral. Moreover, we emphasize the need for platform governance regulation that shifts the burden of proof for regulatory compliance from regulators to platform owners.

Keywords Digital platform governance · Digital platform regulation · Digital competition · Qualitative research · ICTs and society · Information systems · Social welfare computing

JEL classification L12 · L14 · L22 · L4 · L51 · M15 · K21 · O38 · D42

Introduction

Digital platforms, including social media, search engines, marketplaces, and mobile operating systems, have profoundly influenced society and the economy through data-driven business models that focus on efficient matchmaking (Abdelkafi et al., 2019; Bartelheimer et al., 2022). As of 2024, five of the seven most valuable companies—Google (Alphabet), Apple, Meta (formerly Facebook), Amazon, and Microsoft (GAMAM, formerly known as GAFAM)—are based on platform business models (PWC, 2024). Successful platforms rely on elaborate governance design—i.e., "the structure, process, and methods to make decisions related to collective activities concerning the platform" (Mukhopadhyay & Bouwman, 2019, p. 334) — which works to shape platform mechanisms and network effects, such as openness, switching costs, and incentives for ecosystem actors (Armstrong, 2006; De Reuver et al., 2018). In brief, platform governance orchestrates the power dynamics of platform ecosystems and their involved actors (Hurni et al., 2022).

Big digital platforms have evolved beyond their original business ideas into complex conglomerates, fostering new ways to create and capture value across industries (Gawer, 2022; Gleiss et al., 2023). Current expansions into extended reality, spatial computing, and generative artificial intelligence (e.g., Apple Vision Pro, Metaverse, Microsoft OpenAI, Google Gemini) and regulated industries are testament to the growing technological and economic dominance of such platforms (Degen & Teubner, 2024; Ozalp et al., 2022).

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Meanwhile, increasing accusations against these companies of undermining regulation, monopolizing data and customer interfaces, and prioritizing their interests over public welfare highlight how such concentrated power poses threats to the economy, democracy, and society (Clemons et al., 2022a; Gawer & Bonina, 2024; Weinhardt et al., 2024). This raises the question of "whether existing regulatory approaches and instruments are sufficient to promote and safeguard public interests" (Nooren et al., 2018, p.1). It is therefore necessary to identify and challenge potential negative implications before it becomes too late to react (Clemons et al., 2022a).

In practice, regulators worldwide are only just beginning to implement measures to address the dominance of digital platforms and their negative impact on competition, as seen with initiatives like the EU Digital Markets Act and the US Digital Platform Commission Act (Heimburg & Wiesche, 2023a; Pfeiffer et al., 2024). However, truly effective measures have so far failed to materialize, and competitive advantages remain or increase, prompting calls for more radical regulation—including from Facebook co-founder Chris Hughes who said: "It is time to break up Facebook" (Hughes, 2019, p 1). As a result, the current discourse on digital platform regulation is concerned with not *whether* but *how* platforms should be regulated, and to what extent they should be disintegrated to promote fair competition and public accountability (Kwoka & Valletti, 2021). We identify two major reasons for this shift. First, previous regulations (e.g., GDPR) have targeted the symptoms rather than the broader, dynamic design of platform governance (Gleiss et al., 2023). Second, digital platforms come in various types, with conglomerates often encompassing multiple platform types simultaneously. It is therefore crucial to consider the specific type of platform when applying measures to strengthen competition, as one-size-fits-all approaches fail to address the complexity and diversity of digital platform markets (Gawer & Bonina, 2024).

The interdisciplinary discourse on digital platform regulation (DPR) is primarily driven by the fields of, e.g., industrial organization, competition law, and digital policy. The Information systems (IS) research domain has extensively explored the innovation- and efficiency-generating characteristics and mechanisms of digital platforms (De Reuver et al., 2018; Hein et al., 2020; Rietveld & Schilling, 2021), but paid comparatively little attention to regulatory dimensions (Van Veldhoven & Vanthienen, 2022; Yoo et al., 2024). Themes around information technology (IT) regulation (Butler et al., 2023) or social welfare computing (Clemons et al., 2022b) have only recently entered the spotlight. This is surprising given that Google's founders already anticipated in 1998 that "advertising-funded search engines will be inherently biased towards the advertisers and away from the needs of the consumers" (Brin & Page, 1998, p. 15). This bias towards monetization for the benefit of platform owners is now at the forefront of the regulatory debate.

Thus, more research is needed to better understand the nature of digital platforms from a regulatory perspective, including the importance of platform types and governance design. Platform governance design determines how platforms orchestrate their ecosystem while at the same time shielding it from the regulator's grasp (Gorwa, 2019), thus sometimes departing from their declared mode of ecosystem governance (Harraca & Gawer, 2023). It is therefore surprising that a systematic analysis of platform types and governance design, specifically focusing on digital competition from a regulatory perspective, is still lacking in IS research (Heimburg & Wiesche, 2023a; Mukhopadhyay & Bouwman, 2019).

Given such urgent practical relevance, this paper explores digital platform governance design and its implications for competition from a regulatory view. In order to answer the research question "How does platform governance impact digital competition from a regulatory perspective?," we use a decomposition approach to identify and systemize platform-governance standard types, before conducting a qualitative meta-analysis of 87 cases with platform-governance problems consisting of regulatory decisions and investigations from the USA and Europe.

Our results are threefold. First, following Nickerson et al. (2013), we developed a taxonomy of governance-induced competition problems based on specific governance strategies. Second, by applying this taxonomy to the empirical cases via morphological analysis, we identified four platform-governance standard types from a regulatory perspective. Third, we theorize the interplay between platform-governance design, competition problems, and potential regulatory remedies, aiming to bridge the gap between the IS focus on platform governance and the broader regulatory discourse.

The paper is structured as follows: The next section presents the foundations of digital platforms and the interplay between governance design and competition problems. "Background" section describes our methodological approach. "Towards platform-governance standard types from a regulatory perspective" section outlines our taxonomy and the four identified platform-governance standard types. "The role of platform governance abuse in the rise of hybrid-conglomerate platforms" section discusses the interplay between platform governance, competition problems, and necessary regulatory interventions, focusing on hybrid-conglomerate platforms and offering recommendations for regulators. Finally, "Conclusion" section concludes our study, reflecting on its limitations and future research directions.

Background

Digital platforms, governance design, and regulation theory

Digital platform ecosystems are crucial in our interconnected digital world (Hein et al., 2020). The reduction of search and transaction costs makes them effective matchmakers that orchestrate multiple actor groups to enable shared value creation (De Reuver et al., 2018; Rochet & Tirole, 2003). Broadly speaking, a platform can be described as a modularized architecture with a reliable design that controls how its ecosystem actors and components interact (Baldwin et al., 2009).

A platform ecosystem comprises some key components: platform owners who hold varying degrees of centralized power; platform actors, who contribute to and operate autonomously within the ecosystem (De Reuver et al., 2018; Hagiu & Wright, 2015); and platform value-creating mechanisms for facilitating transaction and fostering innovation (Hein et al., 2020). While actors must be affiliated with the platform, owners steer the key terms of transactions (e.g., pricing or quality of traded goods). Interaction and affiliation involve three types of usage and membership externalities: (1) Platform openness is the degree to which an owner allows (potential) actors to access, contribute to, and interact with its core ecosystem functionalities and data. (2) Network effects (direct, indirect, data) arise from the optimal pairing of actors; i.e., a platform's usefulness (and thus value) is determined by the size and fitness of other actors or goods. Direct network effects enhance value as the number of same-side actors increases (e.g., users on WhatsApp), while indirect network effects generate value through actors' growth on the complementary side of the market (e.g., buyers and sellers on Amazon marketplace). Data network effects occur when the value of a product or service increases through the generation, collection, and effective utilization of interconnected data within a network (Gregory et al., 2022). (3) Homing and switching costs bind actors to the platform through (perceived) lock-in effects or comparative benefits. (Evans & Schmalensee, 2005; Rochet & Tirole, 2003).

Platforms also vary based on their purpose and application. From a technical perspective, digital platforms are innovation platforms that "serve as a technological building block on top of which innovators can develop complementary products or services (e.g., iOS, Google Android, Linux)" (Gawer, 2022, p. 113). From an economic perspective, digital platforms like Uber or eBay are viewed as transaction platforms that efficiently pair up complementary actor groups (Schreieck et al., 2016). Platforms that comply with both definitions, such as iOS and its App Store, are integrated-hybrid platforms (Abdelkafi et al., 2019). These platforms combine multiple platform types within a single ecosystem and offer open interfaces to potentially unlimited external capabilities (Bonina et al., 2021; Fürstenau et al., 2023). Large platform ecosystems like Apple or Amazon are described as hybrid-conglomerate platforms due to the growing horizontal and vertical integration of various independent platform types and weakly related market segments by a platform owner (Bourreau et al., 2018; Cusumano et al., 2019). Consequently, platform owners rely on efficiently governing their ecosystem and mechanisms to generate and capture value, thereby increasing their market dominance.

Platform-governance design is crucial for governing the ecosystem and is influenced by architecture decisions (Gorwa, 2019; Hanseth & Modol, 2021). Platform governance "refers to the structure, process, and methods to make decisions related to collective activities concerning the platform" (Mukhopadhyay & Bouwman, 2019, p. 334); i.e., the rules that foster or limit mechanisms and network effects (Armstrong, 2006; De Reuver et al., 2018), including aspects of platform openness, homing and switching costs, or financial conditions and incentives for ecosystem actors (Halckenhaeusser et al., 2020; Mukhopadhyay & Bouwman, 2019). According to (Chen et al., 2022), platform governance mechanisms can be split into incentive and control, where incentive mechanisms govern resource sharing or information provision, and control mechanisms govern access or the actors' output. Therefore, governance design can work to turn platform owners into quasi-regulators (Eisenmann et al., 2011).

Given the economic and societal impact of digital platforms and the power vested in governance design to control resources and limit the autonomy of actors, concerns about negative effects (e.g., surveillance, monopoly) have risen with the emergence of dominant platform owners such as GAMAM. These conglomerate platforms increasingly gatekeep the digital world's technical, economic, and social interactions (Clemons et al., 2022b; Van Dijck, 2021a; Zuboff, 2015). DPR is the legal and policy framework governing digital platforms' activities, behaviors, and business practices (Nooren et al., 2018). It comprises various regulatory objectives, including reducing information asymmetries, unequal bargaining power, and anti-competitive behavior (Baldwin et al., 2011). In addition, DPR targets critical resources such as data (Fast et al., 2023), revenue distribution (e.g., taxation), and value creation (Bourreau et al., 2018) and steers platform behavior in specific market segments and situations (Kokshagina et al., 2022; Lindenmayr et al., 2022).

Regulation theory broadly distinguishes between command-and-control and principle-based regulation (Baldwin et al., 2011). Command-and-control regulation relies on

prescriptive rules and standards that mandate specific actions for organizations, ensuring compliance through enforcement mechanisms (Butler et al., 2023). In contrast, principle-based regulation emphasizes goals, affording organizations with greater flexibility and adaptability on how they achieve them (Nooren et al., 2018). Developing effective regulatory frameworks for digital platforms along certain principles is crucial, given their significant economic and social impact (Gawer, 2022; Heimburg & Wiesche, 2023b). Consequently, DPR is perceived as a process marked by collective action challenges that overshadow particular interests (Gleiss et al., 2023).

Research and practice show that problems caused by digital platforms can be divided into platform-related and competition (/monopoly)-related problems (Gleiss et al., 2023). Platform-related regulation problems mostly arise from behavior and interactions among platform owners, actors, and regulators that enable platform owners to play off their strengths or exploit external deficiencies. Sharing-economy platforms like Airbnb or Uber have used such regulatory shortcomings to scale their businesses by ignoring industry-specific regulations that apply to hotels or cab operators (Han et al., 2022; Van Dijck, 2021b). Furthermore, digital platforms conduct regulatory breaches (e.g., of data protection rules) because the strategic value (of data) is higher than (potential) fines (Gleiss et al., 2023). Copyright infringements due to illegal content distribution or moderation issues are further platform-specific problems (Gorwa et al., 2020). Insufficient or excessive content moderation can restrict freedom of expression, enable targeted exploitation for misinformation, and erode public trust (Lindenmayr et al., 2022). Competition-related regulation problems arise from unequal power distribution between platforms, actors, and competitors, often leading to anti-competitive behavior. Platforms with significant market power can arbitrarily set transaction fees (Halckenhaeusser et al., 2020), control access to essential technical interfaces or data (Fast et al., 2023), and restrict actors' ability to switch to competing platforms (i.e., restrict multihoming) (Parker & Alstyne, 2008). In extreme cases, platform owners can entirely capture margins from actors, creating an imbalance in value creation and revenue shares. This imbalance is exacerbated by unequal bargaining power or information asymmetries that favor the platform owners (Gleiss et al., 2023). Furthermore, digital platforms must evolve to increase actor relevance and market share (Eisenmann et al., 2011). The platform envelopment lifecycle proposes that digital platforms progress through various phases—birth, expansion, leadership, and renewal—each necessitating different capabilities and strategies in relation to actors and competitors (Teece, 2017). Given these considerations, three competition-related problems can be identified from a regulatory perspective:

1) Platform owners govern the access and behavior of different actors within a platform ecosystem (Mukhopadhyay & Bouwman, 2019). They define technical, organizational, and financial rules and incentives, such as transaction fees or the extent to which actors can interact (Schreieck et al., 2016).
2) Besides their role as ecosystem orchestrators, some platform owners also behave as actors (e.g., Amazon Marketplace) (Harraca & Gawer, 2023). This dual role enables anti-competitive behavior, since undue preferences based on information asymmetries disadvantage actors that compete with the platform owners (Khan, 2019).
3) When a platform owner seeks to expand its ecosystem horizontally or vertically, it must attract new actor groups from competing ecosystems, leading to direct competition between platform owners. Such direct competition between platform owners strengthens the market power of platform actors. Accordingly, some platform owners have jointly entered into strategic cooperation to protect their ecosystems (Petit, 2020). An example is default Google Search installment on iOS devices, for which Google pays Apple several billion euros annually (Wakabayashi & Nicas, 2020). This prevents Apple from competing with Google Search, but consequently other search engine providers suffer a great disadvantage. Competition among platform owners usually originates from vertical or horizontal market integration through service bundling or expansion to attract actors from other ecosystems while protecting their own actor relations to decrease competitive pressure (Gleiss et al., 2023).

Hybrid-conglomerate platforms can thus purposefully leverage mechanisms to centrally orchestrate platform governance by bundling various innovation and transaction services and establishing strategic cooperation agreements with competitors. The horizontal and vertical integration of hardware, operating systems, and service layers enables platform owners to monopolize user and data ownership (Clemons et al., 2022a; Gleiss et al., 2023). As a result, strong lock-in effects for actors and high market-entry barriers for competitors are created. Therefore, from a competition-policy perspective, despite their attractiveness and generated value for society, hybrid-conglomerate platforms tend to undermine a fair economic playing field, reduce innovation pressure, and diminish consumer welfare orientation and data sovereignty (Lancieri & Sakowski, 2021; von Scherenberg et al., 2024).

Consequently, competition intensity in digital platform markets decreases, leading to negative societal implications

(Cabral et al., 2021; Kumar & Gupta, 2023). These include data exploitation, surveillance, innovation gatekeeping, and the undermining of digital sovereignty (Pohle & Thiel, 2020; Rowe & Markus, 2022; von Scherenberg et al., 2024). To limit these negative impacts, regulators and decision-makers worldwide strive to establish rules for fair competition in digital markets (Pfeiffer et al., 2024). These rules aim to increase the number of direct competitors to mitigate monopoly tendencies arising from network effects, promote innovation, enhance consumer welfare, and enable efficient value creation (Lancieri & Sakowski, 2021).

Digital platform governance and regulation in IS research

The IS community has extensively researched digital platform ecosystems and their implications from technical, socio-technical, and economic viewpoints (Costabile, 2024; Hein et al., 2020). Much attention has been paid to underlying platform mechanisms, relationships between platform owners and actors, ecosystem evolution, and technical and strategic implications (De Reuver et al., 2018; Halckenhaeusser et al., 2020; Schreieck et al., 2016). However, current IS platform governance research predominantly focuses on the perspective of platform owners and is only gradually expanding to include actors and third parties (Heimburg & Wiesche, 2022). For example, Floetgen et al. (2023) highlight the impact of platform-ownership structure on platform governance design, while Hermes et al. (2020b) conceptualize a platform market-dominance framework based on insights into platform envelopment, openness, and governance design. Gawer and Bonina (2024) examine the competitive risks posed by digital platforms in developing countries and recommend that policymakers "distinguish between transaction and hybrid platforms to enhance the focus of antitrust investigations" (p. 1). Despite these advancements, a systematic regulator's perspective on platform governance and digital platforms has been so far neglected in IS research.

This is surprising for three reasons. First, platform regulation aims to erode digital platforms' underlying technical, socio-technical, and economic mechanisms and thus has the potential to reshape existing IS knowledge about digital platforms and governance design (Heimburg & Wiesche, 2023b). Second, researchers from, e.g., industrial organizations, media studies, and digital policy communities have been intensively studying aspects of platform regulation for many years (e.g., Jacobides & Lianos, 2021; Nooren et al., 2018; Van Dijck, 2021a), but the phenomenon of DPR remains under-researched in IS, despite numerous calls for papers and the fact that the topic should—given its position at the confluence of technology, economy, and society—be a focal point for the field (Butler et al., 2023; Pfeiffer et al., 2024; Rowe & Markus, 2022). Third, regulators worldwide are implementing measures to regulate digital platforms and shape markets in the public interest. For example, Heimburg and Wiesche (2023b) have recognized the importance of the EU Digital Markets Act's regulatory discourse for digital platforms and IS research. The same applies to regulating emerging technologies (e.g., AI algorithms) that are frequently utilized by digital platforms and central for policymakers (Kokshagina et al., 2022). Although direct contributions to DPR are rare within IS research (e.g., Butler et al., 2023; Heimburg & Wiesche, 2023a), the discipline has indirectly significantly contributed to DPR by providing broad knowledge of the technical and economic nature of digital platform governance. This knowledge is crucial for regulators to be able to address concerns about the architectural openness of digital platforms, the exploitation of data and algorithms, and their value creation and capture.

Architectural openness describes the openness of the core technological platform to actors and the accessibility of key resources such as interfaces and boundary resources (Hermes et al., 2020b). Therefore, the degree of openness reflects a trade-off between retaining and relinquishing control over a platform, making it a critical governance decision for the platform owner (Benlian et al., 2015). Depending on the architecture modularity, the degree of openness can vary based on the platform's lifecycle phase and the owner's interests (Teece, 2017). From a regulatory perspective, a platform with a large actor base that reduces openness and thus restricts the actors' autonomy should be viewed critically. In this context, multihoming for actors reduces dependency on a platform owner through interoperability, thereby positively affecting competition (Chen et al., 2022).

Data and algorithms are crucial resources and technological artifacts that digital platforms use to generate and capture value. Data and algorithms impact competition mechanisms and the privacy of actors. They can lead to biases in automated decision-making and form the foundation for surveillance capitalism (Kokshagina et al., 2022; Zuboff, 2015). Data plays a decisive role in the sovereignty considerations of platforms and actors, highlighting the significance of the data sovereignty concept (Abbas et al., 2024; von Scherenberg et al., 2024). Data regulations should protect actors against risks and adverse effects stemming from information asymmetries (Trzaskowski, 2022) and empower actors through measures such as data sharing (Krämer & Shekhar, 2024). Data is also crucial for competition between platforms and actors. From a competition policy perspective, the freer the data exchange, the more dynamic and fairer the competition. However, due to underlying platform mechanisms and competitive relevance, platform owners are interested in gaining exclusive data usage control. Regulatory measures in response to this might include mandatory data sharing (Krämer & Schnurr, 2018) or enforcing data

neutrality to prevent platform owners' gatekeeping (Easley et al., 2018). Fast et al. (2023) determine three remedies to address market power and enhance competition in platform markets: (i) empowering consumers; (ii) promoting data openness; and (iii) limiting data scale. Krämer and Shekhar (2024) assess the effectiveness of data-sharing and siloing obligations mandated by the EU Digital Markets Act, concluding that the competition policy should promote data-sharing while avoiding the creation of data silos.

Value creation and value capture are the focus of platform strategy and governance design research. Hermes et al. (2020b) document the interrelations between environmental factors in platform ecosystems, firm-level strategies, and their effects on market dominance. Hermes et al. (2020a) showcase the causes of US platform dominance and Europe's "platform gap," resulting from a fragmented regulatory market, lack of funding, and (technology) dependencies. The vertical integration of digital platforms, from hardware offerings to the operating system level and platform services layer, enables self-preferencing and cross-subsidized market entries (Gleiss et al., 2023). This phenomenon is particularly evident with the market entry of hybrid conglomerate platforms like Google and Apple into regulated industries, such as healthcare and digital identities (Degen & Teubner, 2024; Gleiss et al., 2021). From a regulatory view, a core problem lies in the dual role of the platform owner as orchestrator and actor within its ecosystem. Independent actors perceive the platform owner both as an ecosystem regulator and a market participant, opening the door to anti-competitive platform behavior. Cowls et al. (2023) argue that given the gatekeeping role of app stores and operating systems, a regulatory focus on governance design is critical to enable fair conditions for platform owners and actors. The regulatory problems that result from vertical integration are known from traditional network industries, such as telecommunications markets (Montero & Finger, 2021). For example, Clemons and Madhani (2010) explore whether the antitrust remedies enforced on AT&T through the 1913 Kingsbury Commitment, which led to the establishment of the first government-regulated private monopoly, should be applied to Google. Krämer and Schnurr (2018) investigate the need for neutrality guidelines known from the net neutrality debate in telecommunication markets and conclude that new transparency rules for dominant platforms are needed to identify misconduct and strengthen legal enforcement. The case of BankID (a Finnish digital identity platform) shows that regulation can weaken a platform's market dominance and strengthen competition (Bazarhanova et al., 2020).

Having summarized the academic literature on digital platform governance and regulation, we identify the following six research gaps:

First, in IS research, platform governance is primarily examined and conceptualized from the platform owner's perspective (Heimburg & Wiesche, 2022). For example, it has been suggested that platform owners should aim to create lock-in effects for actors and capture value through platform envelopment to strengthen their market position (Eisenmann et al., 2011; Hermes et al., 2020b). However, platform owners often face public criticism for the negative impacts their governance decisions have on actors, competitors, and society (Lindenmayr et al., 2022). This predominantly focus on the owner's perspective, and to a lesser extent on that of actors, overlooks the regulatory angle, thereby creating a significant research gap in understanding the broader systematic effects of platform governance design.

Second, IS research on digital platforms has primarily focused on individual platforms, without exploring the increasing complexity of hybrid platform configurations (Schreieck et al., 2024). Given the practical realities, it is vital to understand how governance practices vary across different platform types and how these variations underscore the necessity for regulatory differentiation. Our contribution, which concentrates specifically on hybrid platform types, advances IS research by shifting the focus from single-platform phenomena to the dynamics of hybrid platforms from the perspective of the regulator.

Third, research in IS and management has extensively examined and conceptualized the interplay between architecture and governance decisions, emphasizing their interwoven nature in designing thriving platform ecosystems (e.g. Hanseth & Modol, 2021; Jovanovic et al., 2022). However, in light of the increasing regulatory efforts of various authorities worldwide, it is essential to extend the perspective and investigate how governance and regulatory interventions interact to reshape the design of IT architectures, organizations, and ecosystems.

Fourth, recent antitrust research shows that platform types play a key role from a regulatory standpoint, meaning that differentiation and more research are critical given the limited resources of enforcement authorities (Gawer & Bonina, 2024). To address this gap, a systematic analysis of the interplay between governance design and platform type is needed, especially from the perspectives of the USA and EU, where the most powerful platforms operate and the most advanced regulatory discussions and legislation occur.

Fifth, the interdisciplinary platform regulation debate is increasingly concentrating on how hybrid-conglomerate platforms like GAMAM should be regulated and whether they should be broken up or at least disintegrated to increase competition (Heimburg & Wiesche, 2023a; Kwoka & Valletti, 2021). Beyond assessing the effectiveness of such an approach, there is a growing need to explore how these measures can be (theoretically) implemented and which

criteria should guide the enforcement (Fast et al., 2023; Gleiss et al., 2023; Krämer & Shekhar, 2024).

Sixth, expanding the conceptualization of digital platforms to include the interplay between platform owners, actors, and regulators from a public interest perspective can help IS address the societal grand challenges of our time (Wolff et al., 2022). This need is highlighted in recent calls for papers within the IS community that aim to strengthen digital democracy (Weinhardt et al., 2024), directing technology to "minimize societal disruption" (Clemons et al., 2022b, p. 1), and examining the negative externalities of digital (platform) innovation (Yoo et al., 2024).

Methodological approach

Before initiating data collection, we conducted a literature review to identify key issues relevant to our research inquiry. This approach ensures that our study is firmly grounded in the relevant academic literature. We employed dual search term combinations using the keywords "digital platform," "platform governance," "platform regulation," "digital competition," and "data regulation" through Google Scholar and the AISEL database, screening the first 15 results for each combination. Additionally, we performed backward and forward searches to uncover further academic literature in IS, industrial organization, competition law, management and media studies, as well as publications dealing with digital policy themes. This process yielded 82 publications in journals and conference proceedings, which serve as the foundation for the background section of our research inquiry to explore the impact of platform governance on digital competition from a regulatory perspective.

Building on the above-mentioned literature review, we applied qualitative meta-analysis to generate insights and theory through aggregating, interpreting, synthesizing, and re-applying knowledge from empirical findings that are dispersed across time and publications (Hoon, 2013). Qualitative meta-analysis has become more relevant in IS research for developing a theory of complex phenomena by generating evidence-based but generalizable insights into underlying interrelationships (Combs et al., 2019; Hansen et al., 2022; Skinner et al., 2022). We followed Habersang and Reihlen's (2024) recommendations for a qualitative meta-analysis by conducting an exploratory inquiry using a comprehensive search approach for generating high data heterogeneity. On the basis of 87 cases, we essentially followed Nickerson et al.'s (2013) taxonomy development method to decompose the nature of digital platform governance from a regulatory viewpoint and derive standard types through taxonomy application. This consisted of three major steps:

Data collection and sample

We first collected an appropriate collection of suitable cases for our research goal. Given the omnipresence of US platforms in both research (Lindenmayr et al., 2022) and practice (Maeker, 2024), we first analyzed and identified 40 cases from the US Congress report "Investigation of competition in digital markets—majority staff report and recommendations" on the investigations into Google, Apple, Amazon, and Facebook and their implications on competition in digital markets (Nadler & Cicilline, 2022). This comprehensive 370-page regulatory investigation report (part 1—excluding the appendix of more than 2600 pages) synthesizes an extensive collection of 1,287,997 documents and communications. It includes testimonies from 38 witnesses, hearing records spanning over 1800 pages, and 38 submissions from 60 antitrust experts representing diverse political viewpoints. Additionally, the report incorporates interviews with over 240 market participants, former employees of the investigated platforms (Google, Apple, Amazon, and Facebook), and other individuals, amounting to thousands of hours of input (Nadler & Cicilline, 2022).

This report was not only helpful in gathering a valid and objectively curated set of verifiable cases with platform governance competition problems but also in uncovering different facets of the platform conglomerates in view of our decomposition strategy. Using a decomposition perspective, we aim to disintegrate complex platform conglomerates such as GAMAM and thereby shed light on the problem-causing platform types and underlying mechanisms and interdependencies among platform owners, actors, and regulators. Decomposition is a well-established approach for breaking down a complex structure (here, the platform conglomerate) into smaller objects (single platform services) to effectively study the research phenomenon (platform governance from a regulatory view) and generate new insights (Alexander, 1964; Bartelheimer et al., 2022; Wand et al., 1989). For example, our study classifies the Google and Amazon ecosystems as multiple individual platform-service cases (e.g., marketplace, AWS) to reveal the underlying mechanisms and interdependencies of the interwoven structures.

In order to cover all GAMAM organizations, we complemented the 40 cases from the US Congress report with nine additional Microsoft cases. To incorporate the European perspective and broaden the range of platform types (thereby enhancing the generalizability of our findings), we carefully selected a comparable number of additional cases that meet these geographical and diversity criteria. These cases were drawn from regulatory decisions by the European Commission and Member State regulators, as well as from policy and newspaper articles. This triangulation from diverse sources increases the validity, credibility, and completeness of our study (Given, 2008).

Altogether, we collected 87 cases to reach a point of theoretical saturation; i.e., where adding more cases would not generate new knowledge for our research question. Our data sample consists of 45 cases from the USA and 42 from Europe, of which we characterize 34 as regulatory decisions (e.g., the EU Commission investigation of Microsoft Internet Explorer's competition effects in 2009) and 53 as regulatory investigations (e.g., US Congress report or the ongoing EU Digital Market Acts or Services Act investigations). Building on Bartelheimer et al.'s (2022) digital platform systematization, our data sample encompasses a variety of platform concepts needed for robust generalization, including e-commerce platforms, social platforms, sharing platforms, and software and information technology platforms.

Taxonomy development

In the second step, we followed the taxonomy development approach as proposed by Nickerson et al. (2013), with governance-induced competition problems as the meta-characteristic. We first used a conceptual-to-empirical approach to derive and develop a subset of taxonomy characteristics and instances from the literature before employing an empirical-to-conceptual approach; i.e., we contrasted and aligned the literature-based taxonomy draft with the case data. By continuing to alternate between employing the literature and cases, we challenged this taxonomy draft through four further iterations until we achieved a stable and comprehensive taxonomy. To increase the quality of our taxonomy, the sixth iteration (empirical-to-conceptual) was conducted by two coders independently, resulting in an intercoder-reliability (ICR) of 0.89, before deviations were discussed until consensus (O'Connor & Joffe, 2020). In order to systematize the characteristics and instances, we applied the morphological method for taxonomy development and representation (Heinbach et al., 2022; Schoormann et al., 2023), a highly systematic approach for structuring multi-dimensional problems. The aggregate of all characteristics and instances is represented by a morphological box, which allows for a structured morphological analysis, systematization, and comparison of complex phenomena at a later stage (Ritchey, 2006, 2018; Zwicky, 1966). Hence, it is very helpful to systematically compare cases and derive standard types.

Standard type derivation

In the final step, we applied the developed taxonomy to the 87 cases in order to derive and describe standard types from revealed patterns. Two researchers independently assigned the appropriate instance of each characteristic in the morphological box for each case with an ICR of 0.81, before deviations were discussed with a third independent researcher until consensus was reached. Finally, the patterns let us determine four governance standard types of digital platforms from a regulatory perspective. The appendix contains the final aggregated analysis, providing an overview of the 87 identified cases and the morphological analysis that led to the standard types. In light of the existing literature and the case contents and backgrounds, we provide some explanation of the interplay of these four standard types, focusing on competition problems. In addition, we assess the interplay of platform governance design and competition problems from a regulatory perspective, propose measures to counter negative impacts, and theorize the relationship between regulators and platform ecosystems from an IS perspective.

Towards platform-governance standard types from a regulatory perspective

Development of the governance-induced competition problem taxonomy

The taxonomy of governance-induced competition problems (presented as a morphological box in Fig. 1) encompasses the evaluation of platform governance and its competitive implications from a regulatory standpoint. It was developed

Characteristics	Instances			
Platform type	transaction	innovation	Integrated hybrid	hybrid-conglomerate
Platform core markets	single core-market		multiple core-markets	
Platform lifecycle	birth	expansion	leadership	self-renewal
Platform-governance strategy	open + neutral	closed + neutral	open + self-preferencing	closed + self-preferencing
Competition-related problem	skim-off effects	infrastructure	undue privileging	cross-subsidization
Platform-governance problem	contract design/enforcement	participation	platform activity	resources
Exploiting party	actor		platform	
Disadvantaged party	actor + society	market competitor	actor competitor	platform competitor
Remedy	setting requirements	multi-homing for actors	access for actors	sharing with competitors

Fig. 1 Governance-induced competition problem taxonomy

by integrating literature review and empirical data, with the literature highlighting the characteristics platform type, life cycle, governance strategy, and competition-related problems, while the empirical data highlighted platform-governance-induced competition problems, their implications for third parties, and potential remedies.

We identify four platform types, categorized into simple and hybrid platforms, each with two subtypes. Simple platforms are transaction (e.g., booking.com for hotels) or innovation platforms (e.g., Linux) and are active in one core market. Hybrid platform types operate in multiple core markets and can be differentiated into integrated-hybrid platforms and hybrid-conglomerate platforms based on how many platform services are bundled. An integrated-hybrid platform interlocks adjacent single platforms and comprises at least two distinct innovation or transaction platforms (Gawer, 2022). In contrast, a hybrid-conglomerate platform arises when a central governance structure coordinates multiple single or integrated hybrid platforms via various horizontal and vertical layers, akin to the organization of large industry conglomerates (Cusumano et al., 2019). There are two kinds of platform core markets. Simple platforms operate within a single core market, while hybrid platforms extend across multiple core markets. The hybrid conglomerate platform integrates the largest number of market segments under a unified ecosystem governance structure, leading to the highest level of market concentration.

Platform lifecycle describes the platform owner's changing strategy towards actors and competitors and is shaped by platform governance design (Teece, 2017; Xu & Wang, 2022). As described by Teece (2017), a platform in the birth phase seeks to rapidly expand its ecosystem by creating attractive conditions for various actor groups to create and capture value from innovation; the expansion phase describes a mature ecosystem where the platform owner aims to monetize value, benefiting from strong network effects that lead to actor dependency and negative implications for competitors; during the leadership phase, a platform owner seeks to strengthen its position and value share within the ecosystem by acting both as actor and platform owner; and in the self-renewal phase, the platform owner aims to introduce new ideas into the ecosystem and attract additional actors from horizontal or vertical ecosystems, thereby expanding its reach into other markets. (Teece, 2017).

From a regulatory perspective, the platform governance strategy is characterized by a combination of openness (Benlian et al., 2015) or closedness and neutrality or self-referencing. Neutrality and openness are beneficial for avoiding competition problems since they reduce platform owners' control over actors and competitors. In contrast, ecosystem closure and self-referencing have negative competitive effects as they enable value capture in favor of the platform owner. The degree of (data) neutrality is particularly important when the platform owner assumes multiple roles within the ecosystem or competes with other platform owners (Easley et al., 2018; Khan, 2019).

Competition-related problems enable the platform owner to dominate the market and have negative implications for actors or competitors. Building on (Gleiss et al., 2023), skim-off effects describe platform owners' unfair behavior against platform actors, which takes advantage of the high dependencies that occur as a result of unequal bargaining power. An infrastructure problem arises when the platform owner's offering results in high switching costs, limiting actors to generating value primarily through the platform owner's services. From a socio-economic perspective, platform owners bind various actors by supplying essential infrastructures (e.g., cloud services) that reinforce their dependency on the platform. Undue privileging takes place when a platform owner's in-house goods or services are preferred due to its dual role as platform owner and actor in the ecosystem and competing platform actors suffer from information asymmetries (Rowe & Markus, 2022). Finally, cross-subsidization involves exploiting existing resources from a core ecosystem (e.g., the platform owner's actors base) to strategically extend the ecosystem into vertical or competing ecosystems. This strategy is enabled by bundling various offerings, which creates a regulatory problem when competition with other platform owners is limited or hindered due to an uneven playing field (Heimburg & Wiesche, 2023b). Additionally, the developed taxonomy contains four platform-governance problems: contract design and enforcement; participation; platform activities; and resources. These can lead to competition-related regulatory problems, while simultaneously increasing the platform owner's value share.

Control over contract design and enforcement allows platform owners to dominate the design, monitoring, and sanctioning of contractual terms in the ecosystem. Examples are Google's lack of negotiating interest in paying news publishers for their previewed content on Google News (Bossio et al., 2022) or the iOS App Store fees (Nadler & Cicilline, 2022, p. 97) that must be paid to gain access to actors. Platform owners' failure to enforce against actors' eCommerce copyright violations or disinformation on social media platforms are further regulatory problems (Lindenmayr et al., 2022).

Control over participation governance affords platform owners the authority to establish and enforce rules regarding how actors can access the ecosystem and how they may be excluded or hindered. This includes content moderation decisions, such as banning users from social media sites like Facebook or X (Myers West, 2018) and the lack of technical interoperability between cloud platform services, which limits users' autonomy due to high switching costs (Nadler & Cicilline, 2022, p. 117). Participation control can thus

lead to regulatory problems, as it influences the autonomy of actors and (potential) competitors.

The governance control over platform activities leads to the platform owner's exclusive use of ecosystem-generated data, which helps optimize the platform's offerings in the actor role. This creates competitive disadvantages for independent platform actors due to information asymmetries. A prominent example of exploiting this dominance is Amazon Marketplace, where Amazon is a platform owner, retail actor, and logistics service provider in one ecosystem (Nadler & Cicilline, 2022). Independent actor competitors have no access to, e.g., transaction data in the ecosystem, and must buy visibility for additional fees. At the same time, Amazon always preferentially highlights its own offers to the demand side. This leads to a situation in which self-preferenced in-house goods and services get a demand boost, while economic risks of failure are reduced since network effects help to scale new user offerings.

The control over resources enables the exclusive use of data, technical interfaces, the existing user base, or financial power by the platform owner to secure and/or expand its ecosystem to the disadvantage of potential (platform) competitors. For platform owners, controlling their own key resources enables ecosystem growth, while the openness of resources from traditional industry players like banks or the government (e.g., Open Data) helps them gain a double competitive advantage to capture value. For example, Apple uses its near-field communication (NFC) interface exclusively and does not make it available to third parties, although this technical functionality would be the basis for numerous third-party offerings that would increase competition with Apple products (Sweney, 2022). Meanwhile, both Google and Apple benefit from Open Banking data to offer their own services (i.e., Pay or Wallet), thereby further extending their ecosystems (De la Mano & Padilla, 2018).

In all cases, such problems yield exploiting parties (platforms or actors) that harm disadvantaged parties (other actors, society, or competitors). Concerning competition-related problems, platform owners' exploitative behavior towards ecosystem actors is key, but other disadvantaged parties include third parties (e.g., substitute traditional offers) outside of the ecosystem, the general public, market competitors (e.g., traditional offerings), competing actors, and platform competitors—a fact which makes salient significant impact of platforms in our digital world.

To address these regulatory problems, our taxonomy consists of four remedies that aim to minimize the emerging competition problems by influencing the governance design. These are as follows: (1) setting regulatory requirements to steer platforms' behavior in certain market segments and situations; (2) strengthening the autonomy of actors by enhancing multihoming options and improving interoperability; (3) granting access for actors to specific ecosystem data; and (4) sharing key resources for competitors, such as technical interfaces, to help create fairer competition.

Four platform-governance standard types from a regulatory perspective

The application of the morphological taxonomy to the 87 cases revealed various patterns that we then consolidated into four platform-governance standard types (PGSTs) from a regulatory perspective. These standard types, as described in this section, detail the interplay between platform governance and competition-related problems, offering a systematic categorization of digital platform ecosystems from a regulatory viewpoint.

Standard type 1 (PGST1: Growth) in Fig. 2 represents a single transaction or innovation platform type that operates in a core market with a focus on scaling. The platform's governance design is open and neutral to attract actors, creating appealing ecosystem conditions. The dominant governance problem of PGST1 is contract design and enforcement. In this scenario, typical platform-related regulatory problems such as spreading misinformation, copyright violations, or undermining sector-specific regulations arise (e.g., case no. 55; no. 72). For example, Airbnb initially bypassed local tourism laws, while Telegram exemplifies weak content

Characteristics	**Instances**			
Platform type	transaction	innovation	Integrated hybrid	hybrid-conglomerate
Platform core markets	single core-market		multiple core-markets	
Platform lifecycle	birth	expansion	leadership	self-renewal
Platform-governance strategy	open + neutral	closed + neutral	open + self-preferencing	closed + self-preferencing
Competition-related problem	skim-off effects	infrastructure	undue privileging	cross-subsidization
Platform-governance problem	contract design/enforcement	participation	platform activity	resources
Exploiting party	actor		platform	
Disadvantaged party	actor + society	market competitor	actor competitor	platform competitor
Remedy	setting requirements	multi-homing for actors	access for actors	sharing with competitors

Fig. 2 Standard type 1 (PGST1: growth)

Characteristics	Instances			
Platform type	transaction	innovation	Integrated hybrid	hybrid-conglomerate
Platform core markets	single core-market		multiple core-markets	
Platform lifecycle	birth	expansion	leadership	self-renewal
Platform-governance strategy	open + neutral	closed + neutral	open + self-preferencing	closed + self-preferencing
Competition-related problem	skim-off effects	infrastructure	undue privileging	cross-subsidization
Platform-governance problem	contract design/enforcement	participation	platform activity	resources
Exploiting party	actor		platform	
Disadvantaged party	actor + society	market competitor	actor competitor	platform competitor
Remedy	setting requirements	multi-homing for actors	access for actors	sharing with competitors

Fig. 3 Standard type 2 (PGST2: consolidation)

moderation and a strategy of minimal legal enforcement due to poor cross-border cooperation, legal loopholes, and ineffective regulatory enforcement (e.g., case no. 62; no. 78). These problems occur because the platform owner's lax enforcement allows one group of actors to exploit platform offerings, potentially harming other actor groups or third parties. Platform governance exploitation occurs both actively from the actor side and passively from the platform-owner side, which profits from the value creation in the form of skim-off effects. Therefore, the remedy to address PGST1 is to set regulatory requirements that steer the platform owner's behavior and governance design to minimize negative effects in the public interest.

Standard type 2 (PGST2: Consolidation) in Fig. 3 represents a transaction platform with the primary objective of securing the platform owner's market position and enhancing their value share. The platform owner's market power, which leads to actor dependency, allows exploitation at the owner's benefit through a closed and neutral governance strategy. Governance decisions that restrict actors' autonomy within the ecosystem or make it difficult for them to switch to other platform offerings (infrastructure-like dependence) and that enable the platform owner to take advantage of this (skim-off effects) further reinforce this advantage. Platform actors rely heavily on the platform's availability as the ecosystem matures, which leads to high commercial and technical barriers to replacing the platform with a competing offering, as often occurs in traditional network industries (e.g., the telecommunication sector) (Montero & Finger, 2021). Commercial barriers that increase switching costs for actors include "most favored" clauses, where platform actors agree to always offer the lowest price on the platform (e.g., case no. 36; no. 76). Such practices—which restrict the multihoming of platform actors—have been investigated by competition authorities attempting to ensure fair competition in digital markets (Heinz, 2016). A technical barrier is the lack of interoperability for platform actors, stemming from governance decisions to use closed standards in, e.g., voice assistant or cloud services segments (e.g., case no. 19; no. 20; no. 28). Implementing technical standards and regulations on data transfer, such as enabling the transfer of the platform actor's reputation from one ecosystem to another, would help decrease these barriers (Hesse & Teubner, 2020). The dominant platform-governance problem is therefore control of participation, where platform actors' multihoming is restricted via the infrastructure-like constraints of the platform, enabled by the dominant market position of the platform.

Standard type 3 (PGST3: Extension) in Fig. 4 describes an integrated-hybrid platform type that adopts an extension strategy to increase value capture from the ecosystem. In this

Characteristics	Instances			
Platform type	transaction	innovation	Integrated hybrid	hybrid-conglomerate
Platform core markets	single core-market		multiple core-markets	
Platform lifecycle	birth	expansion	leadership	self-renewal
Platform-governance strategy	open + neutral	closed + neutral	open + self-preferencing	closed + self-preferencing
Competition-related problem	skim-off effects	infrastructure	undue privileging	cross-subsidization
Platform-governance problem	contract design/enforcement	participation	platform activity	resources
Exploiting party	actor		platform	
Disadvantaged party	actor + society	market competitor	actor competitor	platform competitor
Remedy	setting requirements	multi-homing for actors	access for actors	sharing with competitors

Fig. 4 Standard type 3 (PGST3: extension)

scenario, vertical diversification occurs, with the platform owner also operating as an actor within its ecosystem. This dual role allows the platform owner to be active in multiple market segments, orchestrating both platform owner and actor offerings to maximize value. As a competition-related problem, undue privileging arises because the platform owner can utilize resources like transaction data and default interface settings to favor its actor offerings over independent actors (Nadler & Cicilline, 2022, p. 16; case no. 30; no. 38), creating a competitive advantage enabled by self-preferencing governance design. Platform activities constitute the primary governance problem here, with the platform owner as the exploiting party (who defines which data is collected and who has access to it) and actor competitors as disadvantaged parties. Examples of PGST3 platforms include the Amazon marketplace, the Android and iOS App Stores, and the pre-installation of platform services on new smartphones and operating system versions (e.g., Nadler & Cicilline, 2022, p. 97; case no. 17; no. 31). This scenario is a classic antitrust case of bundling network infrastructure or operating systems with upstream services, similar to what Microsoft did with, e.g., the Windows operating system and pre-installed Internet Explorer and Multimedia player (Economides, 2001; Watzinger et al., 2020; case no. 41; no. 46).

Standard type 4 (PGST4: protect and capture) in Fig. 5 defines a hybrid-conglomerate platform that operates in multiple markets, leveraging network effects via horizontal and vertical bundling and increasing value creation and capture with a closed and self-preferencing governance strategy. In addition to undue privileging, another competition-related problem arises in the form of the cross-subsidization of new offerings through the use of existing financial, technical, and strategic resources, such as the ecosystem's existing user touchpoints, while the primary platform-governance problem is the exclusivity of resources. The Apple ecosystem exemplifies PGST4 because it is characterized by a high degree of closedness and self-preferencing (e.g., case no. 84; no. 87). Regulatory concerns include the lack of open technical interfaces, such as NFC (Sweney, 2022), and restrictive iOS App Store policies, where users cannot sideload applications, and app developers must use Apple's payment system (Nadler & Cicilline, 2022, p. 95; case no. 87).

As in PGST2 and PGST3, the platform owner is the exploiting party in PGST4: protect and capture. In a large ecosystem with many resources that enable significant cross-subsidization, competitors from outside the ecosystem face disadvantages. While this cross-subsidization may lead to short-term economic losses for the hybrid-conglomerate platform, it serves a long-term strategic purpose by potentially eliminating smaller competitors who are forced to adapt their ecosystem conditions to retain their actors. Therefore, hybrid-conglomerate platform owners aim to acquire as many resources (e.g., data, customer touchpoints) as possible, as these are crucial for competitive success due to network effects in digital markets. The strong ecosystem lock-in effects of hybrid-conglomerate platforms result in a large amount of data being stored in silos inaccessible to competitors and third parties (e.g., case no. 20). This situation reduces competitive pressure for hybrid-conglomerate platforms by making it difficult and costly for new competitive offerings—which lack access to these resources—to scale. Consequently, the main competitors of hybrid-conglomerate platforms are other hybrid-conglomerate platforms with similar prerequisites, as illustrated by the market dynamics in the voice assistant segment, where the main providers are Amazon, Google, Microsoft, and Apple (e.g., case no. 20). This effect significantly reduces the number of relevant competitors and explains the high market concentration of platforms like GAMAM in digital markets. When these hybrid-conglomerate platforms engage in anti-competitive behavior (e.g., with exclusive contracts among themselves), they reinforce their own market power and create anti-competitive barriers (Petit, 2020). This situation underscores the regulatory need to implement measures that open up certain key resources (e.g., data, interfaces, infrastructure components) to competitors to create a more level playing field for competition.

Characteristics	**Instances**			
Platform type	transaction	innovation	Integrated hybrid	hybrid-conglomerate
Platform core markets	single core-market		multiple core-markets	
Platform lifecycle	birth	expansion	leadership	self-renewal
Platform-governance strategy	open + neutral	closed + neutral	open + self-preferencing	closed + self-preferencing
Competition-related problem	skim-off effects	infrastructure	undue privileging	cross-subsidization
Platform-governance problem	contract design/enforcement	participation	platform activity	resources
Exploiting party	actor		platform	
Disadvantaged party	actor + society	market competitor	actor competitor	platform competitor
Remedy	setting requirements	multi-homing for actors	access for actors	sharing with competitors

Fig. 5 Standard type 4 (PGST4: protect and capture)

The role of platform governance abuse in the rise of hybrid-conglomerate platforms

Hybrid-conglomerate platforms exhibiting PGST4 (e.g., GAMAM) also benefit from the accumulation and exploitation of the competition-related problems of the previous PGSTs, since, as revealed by our decomposition analysis, their sub-platform service configurations and lifecycle phases are described by PGSTs 1–3. From a regulatory perspective, the lack of intervention and failure to address competition problems arising from platform-governance design in the early stages has enabled the rise and significant market power of hybrid-conglomerate platforms. Figure 6 schematizes the four PGSTs.

Simple platform types (PGSTs 1 and 2) operate in a core market. The core regulatory question here concerns the openness of the platform owner's governance design. Open platform ecosystems (PGST 1) with few governance rules (or lack of enforcement) by the platform owner tolerate abuse by platform actors. Closed ecosystems (PGST 2) utilize a platform governance design that maximizes the value share in the platform owner's favor. Here, governance design decisions (alongside the strong network effects of a mature ecosystem) restrict and complicate actor multihoming and increase actor dependency. From a regulatory perspective, the focus in PGST 2 must be on identifying such practices, empowering platform actors, and increasing interoperability.

Hybrid platform types (PGSTs 3 and 4) are active in multiple markets, so that ensuring platform owners' neutrality towards platform actors and competitors becomes a key regulatory priority. Due to their dual role in one ecosystem, open and self-preferencing platforms (PGST 3) have a competitive advantage that should be reduced by regulatory measures that ensure the platform owner's neutrality towards all platform actors. For example, platform owners can be banned from using ecosystem data or required to share any data that is available to their own actor segment with third-party competitors. The latter is a more economically efficient regulation than a ban or siloing, as sharing resources leads to new efficiencies and innovation within the platform ecosystem (Krämer & Shekhar, 2024). When a platform selectively expands into new segments (PGST 4) by leveraging the platform governance problems from the previous standard types, a strong pull effect for the new actor groups is created through attractive boundary conditions. This strengthens competition between platforms, but also creates competition problems due to power asymmetries. From a regulatory perspective, the challenge lies in differentiating positive competition effects from negative ones that lead to a winner-takes-all spiral, which not only affects actors and small platforms but also larger platforms that are less vertically integrated across different layers. An example is the change of Apple's default privacy settings, which aimed to reduce monitoring by third-party advertising providers (Conger & Chen, 2022). This small change had a strong negative impact on Meta's core business model, which depends on ad tracking and corresponding revenue streams. Apple, Amazon, and Microsoft exemplify how deeply integrated horizontal and vertical ecosystems (of devices, operating systems, and platform services) can significantly enhance value capture through orchestrated shared governance. This interweaving of business segments raises entry barriers for competitors, leading to high market concentration and reduced fair competition.

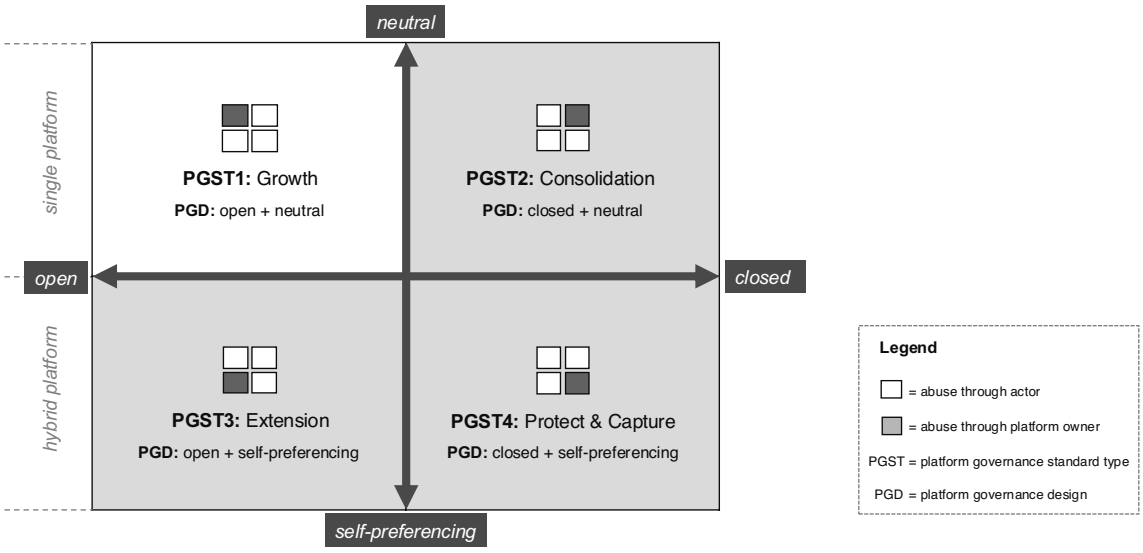

Fig. 6 Schematic representation of the four PGSTs from a regulatory perspective

Policy contribution: implementing remedies to address the market power of hybrid-conglomerate platforms

The governance design of hybrid-conglomerate platforms like GAMAM presents significant competition-related problems that have accumulated as a platform matures through the four PGSTs—assuming that previous competition problems have remained unaddressed by the regulator. Network effects also exacerbate competition problems if they are not addressed. These problems, detrimental to competing actors and platforms, have been aggravated by regulatory inaction. To ensure fair competition, regulatory measures to disintegrate these platforms are necessary. Applying the identified PGST-specific remedies to the sub-platform service configurations of hybrid-conglomerate platforms will effectively mitigate competition-related problems, reduce complexity, and streamline regulatory intervention. Table 1 maps the problems of the four PGSTs and the recommended solutions.

To demonstrate the value of this decomposition approach, we apply it to the hybrid-conglomerate of Google and its various offerings. The video platform YouTube is financed by advertising and exemplifies PGST 1. Since it benefits from advertising revenue by failing to address harmful content distribution, a regulatory debate on content moderation is underway to mitigate negative implications and enforce the public interest by steering the platform's governance decisions. Due to its infrastructure relevance for actors (e.g., case no. 19), Google Cloud represents PGST 2. The offering wields significant market power due to economies of scale, and users face high switching costs because interoperability and multihoming are more challenging. It is therefore essential to strengthen actors' autonomy, which the regulator can promote through technical standards and specifications—as exemplified by the European Gaia-X initiative in the cloud sector or by various data space initiatives (Otto & Jarke, 2019). As a marketplace, the Google Play Store exemplifies PGST 3. Google benefits from transaction data of the app store, gaining economic insights into the success and usage frequency of independent actor offerings. Using this data, Google can develop its own competing products and disadvantage independent actors by restricting their visibility through governance decisions. Finally, Google Pay and Wallet exemplify PGST4, having been launched pre-installed on users' Android OS smartphones and being mandatory for PlayStore users (e.g., case no. 30). Not only did this default access to customers provided Google with a competitive advantage over financial service providers, but the organization further benefitted from the fact that banks were forced to make their customer data available to third party competitors (e.g. Google, FinTechs). This created a double information asymmetry, which constitutes a particular disadvantage for traditional financial service industry competitors (De la

Table 1 Mapping PGSTs' competition problems and remedies

PGST		Competition problem			Solution	
No.	Name	Platform-governance	Competition-related	Disadvantaged party	Remedy	Goal
PGST1	Growth	Contract design & enforcement	Skim-off effects	Actor + society	Setting requirements	Safeguarding the public interest
PGST2	Consolidation	Participation	Infrastructure	Market competitor	Multihoming for actors	Strengthening actors' autonomy
PGST3	Extension	Platform activity	Undue privileging	Competing actor	Access for actors	Strengthening the actor's competitive position
PGST4	Protect & Capture	Resources	Cross-subsidization	Platform competitor	Sharing with competitors	Ensuring a competitive level playing field

Mano & Padilla, 2018). Thus, Google effectively entered the mobile payment and financial services markets by leveraging its existing technical, economic, and strategic resources (Ozalp et al., 2022).

To address regulatory issues, it is crucial to identify competition problem configurations and enforce measures to address them effectively. However, an information asymmetry exists between the platform owners and the regulator, who does not have full access to the platform's internal activities and dynamic governance decisions, making it challenging to identify and address problems. Platform owners further complicate this problem by concealing anti-competitive behavior from the regulator via "two-faced governance" strategies, as highlighted by (Harraca & Gawer, 2023). In order to ensure that remedies can be successfully implemented, the burden of proof for compliance must shift from the regulator to the platform owner. Figure 7 illustrates the current interplay between the platform owner, disadvantaged parties, and the regulator.

Our study reveals how platform governance design can facilitate exploitation by certain parties—primarily platform owners—ultimately harming third parties and competition dynamics. Since, for platform owners, the primary role of platform governance is to orchestrate value generation, it is not surprising that governance design often generates a competitive advantage for the platform owner, as indicated by our four PGSTs. Consequently, harmed actors, competitors, and the public call for regulatory measures to remedy this. Our analysis identifies four principle-based regulatory remedies that mitigate competition problems through platform governance: (i) setting regulatory requirements to achieve more effective platform governance enforcement; (ii) enhancing actors' autonomy; (iii) providing independent actors access to ecosystem data; and (iv) enforcing sharing of critical resources (e.g., critical IT artifacts) with platform competitors. Given the technical complexity and dynamic nature of digital markets, alongside the lack of incentives for the platform owner, the burden of proof for compliance must reside with platform owners. Shifting the burden of proof allows regulators and the critical public to focus on evaluating the compliance and effectiveness of the implementation.

In the digital age, there is an urgent need to move beyond traditional ex-ante regulatory requirements and ex-post enforcement. Due to the critical role of IT artifacts in such

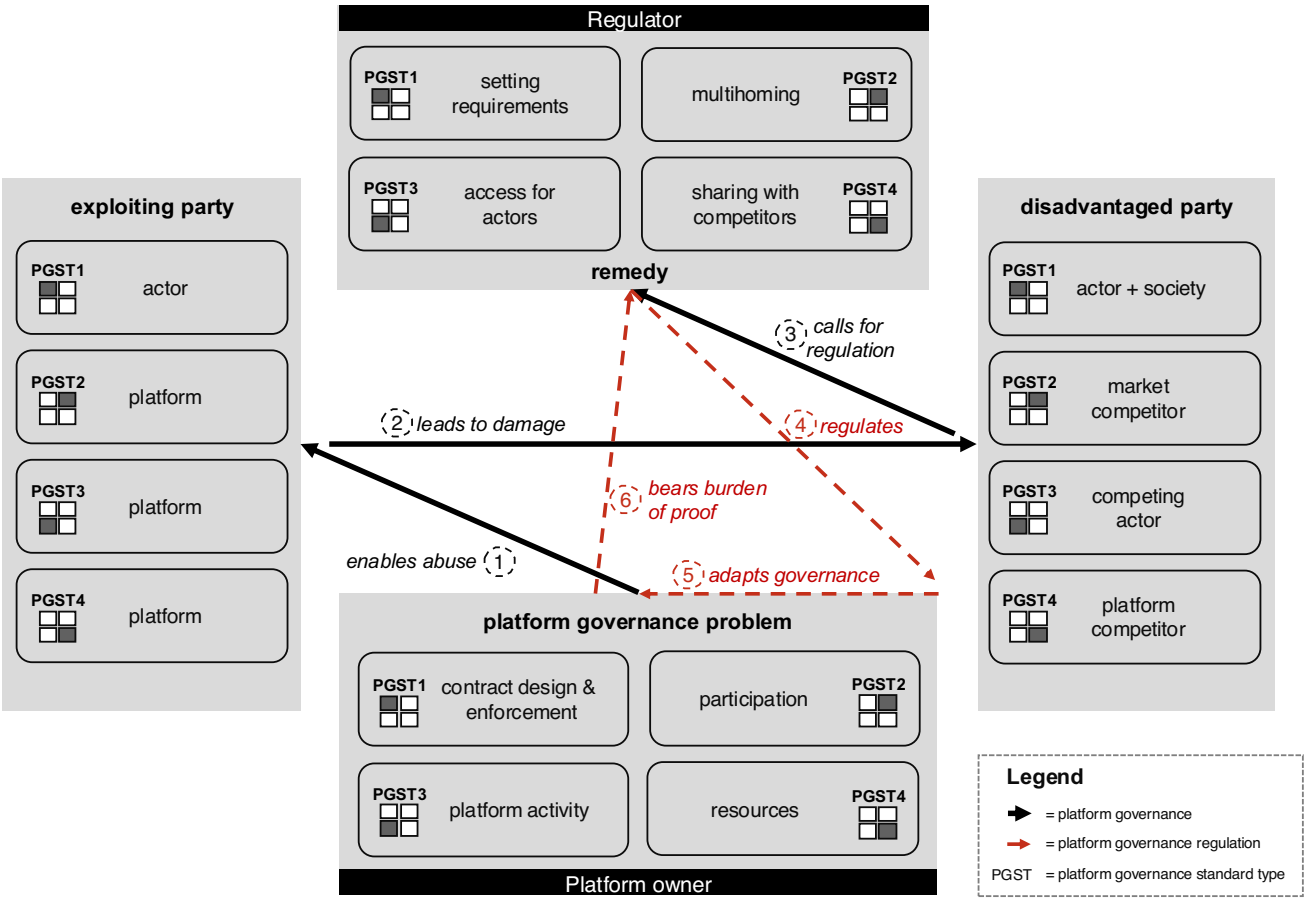

Fig. 7 Interplay between platform owner and regulator to govern digital ecosystems

ecosystems, there is a need for more effective near- or real-time regulation, which requires that regulators are digitally empowered and equipped with the requisite tools to enforce and monitor compliance. For example, technical interfaces, algorithm design, and user-preference settings are small-scale configuration options that enforce the interests of the platform owner through platform governance, making them a critical focus for regulatory attention. Acting as "meta-orchestrators," regulators might collaborate with platform owners to steer the ecosystem governance in the public interest while continuing to enable fair value creation.

Theoretical contribution: conceptualizing platform-governance regulation and recognizing the regulatory importance of IT artifacts

Our study confirms that the absence of platform-governance regulation enabled platforms' high market power (Lindenmayr et al., 2022; Lindman et al., 2023). IS research has extensively investigated governance design and platform lifecycles from the perspective of platform owners. From a regulatory perspective, Teece's (2017) four-stage lifecycle model (birth, expansion, leadership, and self-renewal) translates into our four PGSTs (growth, consolidation, extension, and protect and capture), while the platform-owner behaviors Teece identifies lead to specific competition problems (as shown in Table 2).

By analyzing the interactions of platform owners, actors, and regulators, we enhance the understanding of power dynamics within and between platform ecosystems. From a regulatory perspective, vertical integration of the ecosystem is as important as horizontal expansion, as platform owners interlink different activity layers such as hardware, operating systems, and platform services. We confirm Gleiss et al. (2023) findings that bundling layers, a typical strategy of hybrid-conglomerate platforms, generates high platform dependencies for actors and entry barriers for competitors. Additionally, we provide evidence for Heimburg et al.'s (2023b) observation that platform owners increase their share of value creation through self-preferencing and cross-subsidization, thereby negatively impacting competitors (especially those with less vertical integration).

Our findings provide empirical evidence to support Hildebrandt and Wiewiorra's (2024) recognition that the governance decisions of platform ecosystems today cause similar competition problems as observed in traditional network industries, such as telecommunications. For example, self-preferencing arises when platform owners take on the dual role of actor within their own ecosystem. Due to the dependency of other actors, a double increase in value generation emerges for the platform owner. Cross-subsidization enables the continuous expansion of the ecosystem to new actor groups and increases competition pressure on competing actors and platform owners. Therefore, a key finding of our study is that platform owners' lack of openness and neutrality amplifies competition-related problems, highlighting the need for tailored platform governance regulation. The importance of open platform governance for facilitating interactions with actors to generate value is thoroughly examined in IS research (e.g., Benlian et al., 2015; Schreieck et al., 2016), and we confirm its crucial role in fostering fair competition from a regulatory standpoint. Likewise, our analysis also highlights the importance of neutrality, giving weight to arguments for a neutral obligation for digital platforms (e.g., Easley et al., 2018; Krämer & Schnurr, 2018).

In the twentieth century, antitrust cases in network industries were characterized by the goal of increasing social welfare and lowering the economic and political market power

Table 2 Mapping lifecycle phases from platform owner and regulatory perspective

Lifecycle phase	From platform-owner perspective		From regulatory perspective	
	Name	Behavior	Name	Competition problem
No. 1	Birth	A value proposition is devised to capture value from an innovation	Growth	Negative implications for third parties outside the ecosystem and the public are accepted in order to increase value creation
No. 2	Expansion	Scale and refine the business while closing out rivals	Consolidation	Actor mobility is restricted to prevent them from joining competitors
No. 3	Leadership	Keep customers and partners engaged while maintaining a controlling position within the ecosystem	Extension	Dual role of platform and actor within the ecosystem enhances value capture and leads to a dominant market position
No. 4	Self-renewal	Bring new ideas into the ecosystem	Protect and capture	Platform competitors lack access to critical resources, while the platform owner takes advantage of third-party openness to capture value

of telecommunication companies like AT&T (Wu, 2020), unless they were faced by issues of national security. For example, the U.S. Pentagon actively resisted the breakup of companies like AT&T, being concerned about unintentionally supporting the Soviet Union and jeopardizing the US economic and technological advantage during the Cold War (O'Keefe, 2020). Similar arguments resonate in today's discussions about GAMAM and their regulation: President Donald Trump has recently stressed that weakening the hybrid conglomerate platforms could threaten US strategic interests amid the escalating geopolitical competition with China (Grant & McCabe, 2024). The emerging debate emphasizes the fine line between promoting fair competition and safeguarding national security in an increasingly fragmented and polarized digital world.

It has been suggested that breaking up large hybrid-conglomerate platforms, as it was done to AT&T in 1982 (Gleiss et al., 2023), could solve the current GAMAM competition problems (Grant & McCabe, 2024; Kwoka & Valletti, 2021). However, our study demonstrates that such competition problems arise from distinct platform governance issues rather than from the existence of hybrid-conglomerates as such. We therefore suggest that—rather than resorting to a one-size-fits-all breaking-up approach—the differentiation of the four PGSTs that will enable regulators to effectively respond to the diversity of existing platform ecosystems and the various governance configurations that result in governance-related competition problems.

At this, our findings align with Gawer and Bonina's (2024) conclusions on developing countries, emphasizing that platform types are universally crucial from a regulatory perspective and reinforcing the need for authorities to prioritize the regulation of hybrid-conglomerate platforms. Additionally, our findings suggest that single-type transaction platforms tend to raise more competition concerns than innovation platforms, which is why they form the basis of PGST2. Our study supports the conclusion by Gawer and Bonina (2024) that "two types of platform business models—transaction platforms [in our study, PGST2] and hybrid platforms [in our study, PGST3 + 4]—give rise to distinct competitive concerns and elicit different responses from enforcement authorities" (p. 20). Therefore, researchers and practitioners should systematically apply the four PGSTs and regulatory remedies to platform ecosystems to decompose problem configurations and mitigate competition-related problems. These problems will thereby be resolved bottom-up through market mechanisms, in alignment with the principle-based governance regulation approach recommended by Nooren et al. (2018).

The four regulatory remedies are consistent with existing DPR literature (e.g., Heimburg & Wiesche, 2023b). The regulatory requirements for platform governance are examined within the broader debate on algorithmic content moderation and sharing-economy regulation (Han et al., 2022; Riemer & Peter, 2021). Hybrid-conglomerate platforms present unique challenges by undermining traditional regulations and potentially creating adverse effects for ecosystem actors, competitors, and the general public (Gleiss et al., 2023). These problems arise from the passive role of platform owners, who lack the incentive to proactively address competitive challenges, leading to market failures that inevitably trigger regulatory intervention.

The significance of multihoming and interoperability has been extensively discussed (Heimburg & Wiesche, 2022; Hodapp & Hanelt, 2022), although our empirical data reveal a surprising lack of regulatory implementations that address this. Given the importance of standardization and interoperability in platform ecosystems, particularly for promoting data exchange, the issue is becoming increasingly relevant from a regulatory perspective and should be further explored (Borgogno & Colangelo, 2019; Costabile et al., 2022; Heimburg & Wiesche, 2023a).

Data access, siloing, and sharing remedies are assessed in the literature as a sharp but controversial approach to strengthen competition in digital markets (Fast et al., 2023; Krämer & Schnurr, 2018; Krämer & Shekhar, 2024). Our standard-type remedies confirm Fast et al.'s (2023) findings that data portability (PGST2), mandated data sharing (PGST3), and access obligations (PGST4) are effective regulatory strategies to address data-driven market power. That is to say, the identified remedies are well-known in academic discourse, but we apply them systematically, using our decomposition approach, to empirical data. Doing so sheds light on the interplay between platform governance design and competition problems, leading to a deeper understanding of governance standard types from a regulatory perspective and extending the existing IS knowledge on platform governance.

It is critical for effective regulatory intervention that regulators are on an equal technical level playing field with platforms: i.e., by applying IT and data-driven measures similar to those used by platform owners to govern their ecosystem. Our research confirms the need for regulators to digitally transform themselves to cope with the realities of digital platform ecosystems (Butler et al., 2023). Research into computational antitrust focuses on the fact that regulators must assess changing governance strategies and the role of IT artifacts from an in-depth regulatory perspective and use the latter to implement effective remedies (Schrepel, 2021). IS research must devote more attention to this phenomenon, examining how digital platforms and regulators can interact effectively to govern IT artifacts that are shaped by governance and underlying IT architecture decisions. Our focus on platform governance confirms the crucial role of IT in regulatory enforcement, emphasizing the need for greater consideration of this aspect in the interdisciplinary platform

regulation debate (Fast et al., 2023; Heimburg & Wiesche, 2023a). We concur with Heimburg and Wiesche (2023a) that increased modularity in platform architecture helps to mitigate negative competition implications. Therefore, modularity should be reconsidered from a regulatory perspective in IS research, given that various regulations, such as the EU Digital Markets Act, necessitate reevaluating architectural decisions within platform ecosystems. The four remedies of this study align with the EU Digital Markets Act, aiming to address competition problems while maintaining strong incentives for value creation. Effective platform-governance regulation must strike a balance between fostering value creation and ensuring fair value distribution.

Therefore, it is time to put an end to the situation where platform owners govern their ecosystems with strategies that undermine fair competition, while regulators appear ineffective and powerless. US Senator John Sherman, the architect of the 1890 Sherman Antitrust Act—the world's first antitrust law—underscored the vital need for regulatory intervention to maintain fair competition and a balanced power distribution between public and private interests. His words hold even greater relevance today, as hybrid conglomerate platforms increasingly dominate our networked economy and society. Sherman's statement, "If we will not endure a king as a political power, we should not endure a king over the production, transportation, and sale of any of the necessaries of life" (Sherman, 1890. p. 2457),[1] highlights the need for regulatory intervention. As our study demonstrates, just as the nineteenth and twentieth centuries confronted conglomerate giants (e.g., Standard Oil, AT&T) today's digital platform economy demands robust and tailored-made regulatory remedies to prevent any single organization from monopolizing essential aspects of our digital age—from technology and data resources to markets and societal governance.

Conclusion

The rise of hybrid-conglomerate platforms such as GAMAM has resulted in significant market power concentration and negative competition implications between platform owners, actors, and competing ecosystems. Platform governance design is critical in shaping market dynamics and competition and enabling abuse that favors the platform owner. The complexity of digital ecosystems (i.e., the progression from simple transaction or innovation platform types to hybrid-conglomerate platforms) makes a systematic analysis of platform governance design crucial to addressing negative implications from a regulatory perspective.

To explore how regulators can mitigate these negative competition impacts, we pursued a qualitative meta-analysis to explore, from a regulatory viewpoint, the interplay of platform governance design and competition problems that lead to platform dominance. By developing a platform-induced governance-problem taxonomy and applying it to 87 empirical platform-governance problem cases, we identified four distinct platform-governance standard types (PGST1: growth, PGST2: consolidation, PGST3: extension, and PGST4: protect and capture). To address the identified competition-related problems, platform governance must be more open and neutral towards ecosystem actors and platform competitors. This can be achieved by setting regulatory requirements for active governance enforcement (PGST 1), enhancing actor autonomy through multihoming and interoperability (PGST 2), providing independent actors access to ecosystem data (PGST 3), and enforcing the sharing of critical resources with platform competitors (PGST 4). Such regulatory intervention will be essential to resolving current competition problems in favor of the public interest.

Therefore, our study highlights the necessity of breaking up existing knowledge in digital platform research: First, by emphasizing the importance of platform governance from a regulatory perspective, we address a significant research gap in understanding platform-governance design and shed light on the power dynamics between platform owners, actors, and regulators. Second, we demonstrate that the rise of hybrid-conglomerate platforms has been influenced by both platform-governance design and the failure of regulators to address competition problems early on, benefiting platform owners to the detriment of competitors and the public. Third, we underscore that effective principle-based regulatory intervention needs to respond to the distinct characteristics and problems of each PGST, while shifting the burden of proof for compliance from the regulator to the platform owner. Fourth, we emphasize the necessity of focusing on IT artifacts that implement governance configurations. Regulators must be digitally transformed in order to be able to closely scrutinize these artifacts and be capable of intervening quickly to correct platform owners' dynamic governance decisions. Fifth, given the increasing regulation attempts of digital platform markets, the regulator should be perceived as part of the ecosystem, particularly as hybrid-conglomerate platforms increasingly expand into regulated sectors to capture value (Degen & Teubner, 2024; Ozalp et al., 2022).

As a contribution to research, we provide a comprehensive systematization that helps to understand and further explore the regulatory problems of digital platforms. We focus on platform design as key to controlling resources and constraining the autonomy of actors and competitors. By doing so, we respond to Yoo et al.`s (2024) call to research the negative externalities of digital innovation and theorize

[1] US Senator John Sherman, speaking in support of the legislation before the U.S. Congress on March 21, 1890.

about the interplay between platform owners and regulators to govern ecosystem value creation. As a contribution to practice, we provide an analytical framework of four standard types that help regulators and policymakers systematically and comprehensively address digital platform competition problems. These standard types reduce complexity and facilitate the design of tailored remedies that can prevent or mitigate such problems.

Our study has some limitations. First, our scope is limited to digital platforms and regulatory approaches in Western societies. Second, our analysis focuses on the role of platform governance and the competition dimension of DPR. However, we recognize that other dimensions, such as social polarization resulting from, e.g., content moderation, might be equally important in designing DPR holistically. Third, while we find evidence that platform governance significantly contributes to the emergence of hybrid-conglomerate platforms, we also acknowledge that other anti-competitive factors (e.g., lobbying power) may also contribute to the market power of these organizations. Fourth, and most importantly, our qualitative research design, including deductive and inductive systematization, relies upon the authors' subjective selection, coding, and interpretation. In awareness of this limitation, our findings build on practical evidence reflected by extant theory and research.

Future research should continue to explore the negative implications of digital platforms, emphasizing the role of governance design and the critical importance of IT artifacts in shaping competitive dynamics. At this, external factors that impact and shape digital markets, such as geopolitical competition, deserve greater attention in IS research. The impact increasingly exceeds single regulatory measures and extends to asserting control over the digital space in favor of strategic national interests. Furthermore, we echo the call for papers by Arner et al. (2024) that underlines the need for research on how various digital regulation attempts "will affect the strategic choices of companies, the development and diffusion of innovative technologies, and ultimately the welfare and wellbeing of societies" (p.1). From a social welfare computing perspective, research should aim to facilitate regulation that results in digital platforms that benefit the broader economy and society, not the other way around.

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Data Availability The data supporting the findings beyond Appendix 1 of this study are not publicly accessible but can be obtained from the corresponding author upon request.

Declarations

Competing interests The authors declare that they have no conflict of interest.

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